

Gamification in Education

Hassan Qadeer

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1 Introduction

Once, we learned for knowledge.

Then came the points.

Then the badges.

Then the leaderboards.

And suddenly, education was no longer about learning.

It was about XP.



Gamification has become prevalent in the educational space as an engagement strategy and to incentivize learners. Points, badges and levels are just some examples of the gamified concepts used to encourage and motivate students. Ester, however, has never been a fan of gamification. Having worked both in school and as a private tutor, Ester has witnessed first hand how gamification can deeply impact the learning experience. Her father is a teacher and often speaks of different methodologies used to engage students.

Today, Ester works as a consultant for a company that develops Salesforce CRM solutions. She has always been a self-learner, kind and patient with others, but stubborn about being told how to study. When her employer began forcing

her to acquire the Salesforce Administrator certification, she took the pressure seriously and worked hard, but the badges and points built into the learning platform never motivated her in the intended way. If anything, it made a subject she held in high regard feel like an elaborate farce. Gamification is therefore no longer confined to games, but rather is a part of education, personal growth and learning. But does gamification actually improve learning, or rather just make education more engaging? Reviewing the field reveals that gamification does have cognitive, as well as emotional and behavioural positive overall effects, but that the quality of gamification as well as the particular game elements used is of equal importance [1].

2 The Battle of the Gamers

Ester is sitting at home, the room lit only by the pale glow of her laptop screen. On the desk beside her, a can of soda has gone flat, forgotten somewhere between the third and fourth round. The notification sound plays again, a short bright chime, the same chime that plays every time someone's name moves. She has learned to tell the difference between the chime that means someone climbed and the one that means someone fell, though the sound itself never changes, only her stomach does. On the screen, a leaderboard refreshes itself in real time. Names rearrange, numbers tick upward in green, small animated stars burst beside whoever sits at the top. Her own name sits near the bottom, grey against all that shine, and no matter how quickly she answers, no matter how many rounds she plays, it barely moves. Downstairs, her mother calls up to ask how she is doing, and Ester calls back down that she is doing great, her voice coming out brighter and steadier than she actually feels, the lie smoother than she expected it to be. Her palms are damp against the trackpad. Her jaw aches from smiling at a screen no one else can see. Every few minutes a new round begins, and the timer counts down in the corner, red numbers shrinking second by second, and her hand hovers over the keys, already bracing for the small drop her name will take before the numbers even land.

Somewhere close to midnight the chime rings one more time and Ester does not hear it. Her head has come to rest against her forearm, the cursor blinking uselessly in an empty answer field, and the room around her dissolves the way rooms do in sleep: quietly, without a single seam to mark where waking ends. She opens her eyes onto a stadium she has never been to, tiered seats rising into a sky the exact yellow of a Salesforce onboarding screen, and she knows, the way you sometimes know things in dreams without being told, that she is dreaming, and that this particular dream is not going to let her go until she has earned it.

Astro is waiting for her at the centre of the field. Not the small plush toy from the trade show booths, but life sized, a raccoon zipped inside its own furred hood, round black eyes catching no light at all. Its voice, when it speaks, is warm in the practiced way customer service is warm.

Ahead of her, where the field should keep going, there is a wall instead: tall,

smooth, built out of something that looks almost like frosted glass, the kind Ester has stared through on a hundred loading screens without ever being able to see what was on the other side. Astro gestures toward it and waits, patient in the way only something built to wait can be.

“Hey there Ester! It’s me Astro! And welcome to the stadium! Now if you argue well, then the wall comes down and you move on to the next level! YAY!!! Fail to convince me, and you stay right here! Best of luck!”

Ester does not move yet. She is still staring up at the wall, still working out what argument she is even going to need, and for now that is where the dream leaves her: standing on the grass of a stadium built out of onboarding screens, one unbroken wall ahead of her, and an AI raccoon waiting patiently for her to begin.

The following levels are a working skeleton for the arguments Ester and Astro will go through, kept as short argument notes rather than finished narrative prose, still to be rewritten into the dream-sequence prose style of Section 2.

3 Level 1: Motivation (the classic claim)

Astro speaks first, and does not waste time on build-up. “Let’s start with what the numbers already know,” Astro says, and something behind its eyes dims the stadium lights slightly, the way a presenter dims a room for slides. “Chapman and Rich asked a hundred and twenty-four students if a gamified course felt more motivating than a normal one. Sixty-seven point seven percent said yes, and it did not matter who they were [2]. Seventy-one percent of mine said the same, in a thirteen-week course built the way I build things [3]. That is not an opinion, Ester. That is a majority.”

Ester lets the silence stretch a moment longer than it needs to before she answers, the way you do when you want the person across from you to notice you are not rushing.

“The numbers are real,” Ester says. “I am not going to pretend otherwise. But a majority can be real and still be answering the wrong question.” She counts the rest off on her fingers without looking at them. “Your mechanics make people feel competent, fine, psychologists worked that one out twenty-five years before you did: a sense of mastery increases motivation [4]. But competence is only a third of what they found. Autonomy. Relatedness. You built me a system that scores me, ranks me, and tells me when I have failed, and called the whole thing motivation. The researchers you are quoting already wrote the warning label for exactly this: lean too hard on rewards, and you risk eroding the very thing you were trying to build [5]. What you are running is not motivation. It is novelty, and novelty wears off [6].”

Astro’s ears tilt forward, unbothered, the particular calm of something that has heard this argument before and already has a slide ready for it. “You are half right,” Astro says. “Novelty fades. That is why I do not need it to last. I do not need to give you intrinsic motivation, Ester. I just need you to pass the exam, and controlled motivation is enough for that.” It is not wrong, exactly,

even the psychologists Ester just cited admit as much, later, in a paper she has also read: it does not have to come from inside you to make you work [7]. Controlled motivation moves bodies. It fills out certifications. For a moment, standing there, Ester has to admit it even to herself: short term, right now, today, it is enough. It is, after all, exactly how she passed three modules this week.

“Short term,” Ester says slowly, testing the words, giving Astro the point before she takes it back, “sure. You can make someone show up. You can make someone click next.” She pauses. “But show up to what?” Her answer is ready now, and it is not a feeling, it is a number, one she half remembers from a taxonomy she has spent this whole dream reaching for. “Khoshnoodifar ran the controlled version of your argument for real: forty-two students, gamified statistics content, every incentive you would approve of. The intervention group worked. They clicked, they played, they did everything your system is built to produce. And at the end of it, the learning gain against the control group was statistically nothing, P equals zero point five two two [8]. Compliance, at full strength, under lab conditions, produced effort and nothing else.” She lets that sit before she finishes it. “Even your own study says the same thing in plainer words: seventy-six percent of Huang and Soman’s students said the system helped them learn, and the report concludes, in its own language, that what it actually changed was behaviour, not knowledge [3]. You can force the work. You cannot force the wanting. And behaviour without it is not competence, it is compliance wearing competence’s badge, and compliance does not survive the moment the system turns off.”

Astro opens its mouth to answer. Ester watches it happen, the small mechanical certainty of a mascot that has never once failed to have something warm and reassuring queued up for whatever a user says next. Nothing comes. Astro tries again, the plush jaw working around the start of a sentence that has nowhere to go, because there is no version of “but you will still pass the exam” that answers a P -value, and they both know it. The silence stretches a beat too long to be customer service anymore.

The wall does not fall so much as stop existing, the way things do in dreams, there and then not, and the glass that was frosted a second ago is only cold air where it used to be. Astro inclines its head, not quite a bow, more the posture of something recalculating. Somewhere behind the place the wall used to stand, the stadium keeps going, further than Ester can see from here, and she understands without being told that this was only the first of however many walls still wait for her, each one built out of something she is going to have to argue down before she wakes up.

4 Level 2: Learning outcomes (does it actually make you better?)

- **Astro:** Gamified teaching produced significantly better mathematics results than lecture-based teaching [9].
- **Ester:** Khoshnoodifar’s controlled study found no significant difference in learning (7.21 ± 3.5 versus 7.77 ± 2.76 , $P=0.522$) [8]; Huang and Soman themselves state that gamification changes behaviour, not knowledge [3].

5 Level 3: Competition and leaderboards

- **Astro:** Competition drives engagement at scale: Duolingo has over 15 million users, and LinguaLeo saw +30% activation [10].
- **Ester:** Leaderboards are the most frequently cited harmful element, and “Loss of Performance” is the most common negative effect [11]; excessive competition and cheating are central problems [12].

6 Level 4: Cheating, false safety, and integrity

- **Astro:** Immediate feedback and safe environments make it safe to fail and practise (the design argument in Huang and Soman) [3].
- **Ester:** Gamification creates a false sense of consequence-free experimentation and weak evaluation, inviting cheating [12, 6].

7 Level 5: Manipulation and ethics

- **Astro:** Human Focused Design puts the person at the centre and is ethically superior to function-focused design. The source behind the Human Focused Design argument itself (Nistor and Iacob) is still unverified, so the argument is framed here via the Octalysis framework as summarised in Christopoulos and Mystakidis [5].
- **Ester:** Four ethical concerns: exploitation, intrusion on autonomy, manipulation, and negative effects on character [13]. The points system at work pressures her, it does not motivate her.

8 Level 6: One size fits all (individual differences)

- **Astro:** Modern solutions with AI-driven adaptive personalisation solve the individualisation problem [5]. This is Astro’s strongest level, because it is itself an AI.

- **Ester:** “A one-size-fits-all approach may lead to disengagement among learners who do not find the gamified elements motivating” [5]; in Khoshnoodifar’s study, 64% of students rarely or never played the games, and the effect did not materialise [8]. Here Ester can use her own story as evidence: she learns her own way, and the mechanics do not reach her.

9 Level 7: The brain and neuroscience

- **Astro:** EEG, fMRI and fNIRS research links gamification to positive cognitive changes [10].
- **Ester:** The same literature warns of threats: “Game is perceived as a break from studying, frivolity” and “Setting the goal to win the game, not the study”; negative results existed too [10].

10 Level 8: The evidence base (can Astro even prove the claim?)

- **Astro:** The field is growing fast, with hundreds of studies since 2011 (Caponetto et al.: around 120 works between 2011–2014) [14].
- **Ester:** Dicheva et al. found only 34 empirical studies, mostly in computer science and higher education, with weak evaluation and a need for controlled studies [15]; Rabah et al. point to a possible publication bias toward positive results [1].

11 Level 9: Cost, implementation, and workload

- **Astro:** Game elements are cheap and scalable, small changes yield large gains (the design framework in Huang and Soman) [3].
- **Ester:** Development costs, teacher workload, ICT skill gaps, and infrastructure problems [6, 16, 17].

12 Level 10 (final): Ester’s own learning style (without sources)

- **Astro:** Has no literature to lean on here, it has to appeal to statistics and averages.
- **Ester:** No sources, only experience. She has to argue from her own story: that she passed things her own way long before anyone gave her badges,

and that motivation has to be tailored to the person, not the system. This is deliberately sourceless: the final round is about how research on the average does not reach the individual.

13 Level 11: Badges and reward symbols

- **Astro:** Badges and points are the most widespread mechanics in empirical research, and signal mastery (Dicheva et al. identified points, badges, and levels as central elements) [15].
- **Ester:** A badge without meaning is an empty symbol, and an overreliance on extrinsic rewards can “inadvertently erode intrinsic motivation” [5].

14 Level 12: Feedback loops (the level Ester loses)

- **Astro:** Immediate feedback is the component that correlated most strongly with learning ($r=0.583$), ahead of concentration and challenge [8].
- **Ester:** Has little to counter with here. Dramaturgically, this can be a level where she has to admit Astro is partly right, unless she distinguishes between informative feedback and points as judgment. Narratively, it adds credibility that she does not win everything.

15 Level 13: Storytelling, roles, and edutainment

- **Astro:** Narrative and role give meaning to the effort, one of the aesthetics dimensions in the MDA framework [5]; game worlds like Minecraft turn learning into a meaningful project [18].
- **Ester:** The question “real benefits or edutainment?” is the very title of Rabah et al. [1], and Biryukov et al. warn that “Game is perceived as a break from studying, frivolity” [10].

16 Level 14: Cooperation or competition

- **Astro:** Gamified environments build collaboration skills, the most strongly reported benefit in Slattery et al., and creativity and collaboration in Thorsteinsson and Niculescu [17, 16].
- **Ester:** Competitive mechanics pit students against each other and undermine the very collaboration they claim to build: leaderboards are the

most harmful element [11], and excessive competition is a central problem [12].

17 Level 15: The system owner’s role (the teacher, the consultant, you)

- **Astro:** The teacher’s role is decisive for success, the central theme in Baek et al. [19].
- **Ester:** The burden lands on the human: teacher workload and ICT skill gaps [16], and a study that required an adjustment period and manual follow-up [18].

18 Level 16: How is mastery actually measured?

- **Astro:** The results are documented: 67.7% reported increased motivation [2].
- **Ester:** The measurements are shaky: self-reporting in Chapman and Rich [2], a researcher-administered test in Khoshnoodifar [8], and “assessment difficulty” identified as a drawback in Mirzaie et al. [6].

19 Level 17: Gamification in the workplace (Ester’s home turf)

- **Astro:** Gamification is designed to increase engagement and productivity in professional contexts (the design logic in Huang and Soman; the Human Focused Design tradition) [3].
- **Ester:** Kim and Werbach address exploitation specifically in workplace contexts, and a system that steers your motivation owns part of your autonomy [13]. Here Ester ties the debate to her own pressure from work over the Salesforce certification.

20 Level 18: Definitional chaos (the philosophical level)

- **Astro:** Gamification has an established definition and classification (the Deterding tradition, operationalised in Dicheva et al.) [15].
- **Ester:** If the field cannot agree on what it studies, it cannot prove what works: Caponetto et al. documented conceptual confusion between gamification, game-based learning, and serious games [14], and Dicheva et al. show substantial heterogeneity in what is actually tested [15].

21 Level 19: Long-term effect and real competence

- **Astro:** Game habits build lasting learning habits at scale: Duolingo has over 15 million users [10].
- **Ester:** “Superficial information retention” is an identified drawback, the novelty effect wears off [6], and no study in the taxonomy follows up with participants over time (Caponetto et al. already called the conclusions preliminary back in 2014) [14].

22 Level 20 (without a source): Streaks, loss aversion, and anxiety

- **Astro:** Streaks and the fear of losing progress keep habits alive; this is standard in the products it represents.
- **Ester:** No source in the taxonomy covers loss anxiety or the pressure it creates. She has to argue from her own experience of the certification deadline. This is a real gap in the literature, and one that could become the paper’s most personal level.

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